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Method and system for engagement of a surgical tool with actuators of a tool drive in a surgical robotic system

Abstract

A system and computerized method for detection of engagement of a surgical tool to a tool drive of a robotic arm of a surgical robotic system. The method may include activating an actuator of the tool drive to rotate a drive disk to be mechanically engaged with a tool disk in the surgical tool. One or more motor operating parameters of the actuator that is causing the rotation of the drive disk are monitored while activating the actuator. The method detects when the drive disk becomes mechanically engaged with the tool disk, based on the one or more monitored motor operating parameters. Other embodiments are also described and claimed.

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Background/Summary

FIELD

(1) An embodiment of the invention relates to control units for detecting the successful engagement of a surgical robotic tool with one or more actuators in a surgical robotic arm of a surgical robotic system. Other embodiments are also described.

BACKGROUND

(2) Surgical robotic systems give an operator or user, such as an operating surgeon, the ability to perform one or more actions of a surgical procedure using the surgical robotic system. In the surgical robotic system, a surgical tool or instrument, such as an endoscope, clamps, cutting tools, spreaders, needles, energy emitters, etc., is mechanically coupled to a robot joint of a surgical robotic arm, so that movement or actuation of the robot joint directly causes a rotation, pivoting, or linear movement of a part of the tool (e.g., rotation of an endoscope camera, pivoting of a grasper jaw, or translation of a needle). Once the tool is attached to (e.g., in contact with) a tool drive in the arm, operator commands may cause movements and activate functions of the attached tool, such as closing clamps, adjusting the bend of an endoscope, extending an instrument outside of cannula walls, applying pressure using a clamping tool, as well as other movements and actions.

(3) Due to the varied nature of surgical procedures, different surgical tools or instruments may be selectively attached to the same arm of a surgical robotic system before and during a surgical procedure. In order to avoid equipment malfunctions during a surgical procedure, it is important

that the surgical tool or instrument not only be attached to but also engaged in a mechanical sense to the robot joint of the surgical robotic arm. That is, mechanisms in the surgical tool that impart motion or enable the activation of instrument features (e.g., opening, closing, cutting, applying pressure, etc.), should be mechanically engaged to the actuators that are in the tool drive of the arm of the surgical robotic system, before the surgical tool is in use during the surgical procedure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The embodiments of the invention are illustrated by way of example and not by way of limitation in the figures of the accompanying drawings in which like references indicate similar elements. It should be noted that references to “an” or “one” embodiment of the invention in this disclosure are not necessarily to the same embodiment, and they mean at least one. Also, in the interest of conciseness and reducing the total number of figures, a given figure may be used to illustrate the features of more than one embodiment of the invention, and not all elements in the figure may be required for a given embodiment.
- (2) FIG. 1 is a pictorial view of an example surgical robotic system in an operating arena;
- (3) FIG. 2 is an illustration of a system for detecting engagement of a surgical tool to a tool drive of a surgical robotic arm;
- (4) FIG. 3 is a block diagram showing a surgical tool, a tool drive, and a control unit;
- (5) FIGS. 4A-4C illustrate different states of a tool disk and a drive disk during an engagement process;
- (6) FIG. 5A is a flow diagram illustrating a process performed by a control unit for engaging a surgical tool with a tool drive;
- (7) FIG. 5B is a flow diagram illustrating another process for a control unit detecting engagement of a tool disk to a drive disk based on one or more operating parameters of an actuator that is driving the drive disk;
- (8) FIG. 5C is a flow diagram illustrating another process for a control unit to detect engagement of a surgical tool with a tool drive of a surgical robotic system;
- (9) FIG. 6 depicts a block diagram of a feedback loop; and
- (10) FIG. 7 depicts a block diagram of a controller for use in a feedback loop.

DETAILED DESCRIPTION

(11) Embodiments of an apparatus, system and method for detection of engagement of a detachable surgical robotic tool to a tool drive of a surgical robotic arm of a surgical robotic system are described herein. In the following description numerous specific details are set forth to provide a thorough understanding of the embodiments. One skilled in the relevant art will recognize, however, that the techniques described herein can be practiced without one or more of the specific details, or with other methods, components, materials, etc. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring certain aspects.

(12) Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, the appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics such as those shown in different drawings may be combined in any suitable manner in one or more embodiments.

(13) Referring to FIG. 1, this is a pictorial view of an example surgical robotic system **100** in an operating arena. The surgical robotic system **100** includes a user console **102**, a control tower **103**,

and one or more surgical robotic arms **104** at a surgical platform **105**, e.g., a table, a bed, etc. The surgical robotic system **100** can incorporate any number of devices, tools, or accessories used to perform surgery on a patient **106**. For example, the surgical robotic system **100** may include one or more surgical tools **107** used to perform surgery. The surgical tool **107** may have an end effector at its distal end (also a distal end of the robotic surgical arm **4** to which the surgical tool **107** is attached), for executing a surgical operation such as cutting, grasping, poking, or energy emission. (14) Each surgical tool **107** may be manipulated manually, robotically, or both, during the surgery. For example, the surgical tool **107** may be a tool used to enter, view, or manipulate an internal anatomy of the patient **106**. In an embodiment, the surgical tool **107** is a grasper that can grasp tissue of the patient. The surgical tool **107** may be controlled manually, directly by a hand of a bedside operator **108**; or it may be controlled robotically, via sending electronic commands to actuate movement of the surgical robotic arm **104** to which the surgical tool **107** is attached. The surgical robotic arms **104** are shown as a table-mounted system, but in other configurations the surgical robotic arms **104** may be mounted in a cart, ceiling or sidewall, or in another suitable structural support.

(15) Generally, a remote operator **109** such as a surgeon may use the user console **102** to remotely manipulate the surgical robotic arms **104** and the attached surgical tools **107**, e.g., teleoperation. The user console **102** may be located in the same operating room as the rest of the surgical robotic system **100**, as shown in FIG. **1**. In other environments however, the user console **102** may be located in an adjacent or nearby room, or it may be at a remote location, e.g., in a different building, city, or country. The user console **102** may comprise a seat **110**, foot-operated controls **113**, one or more handheld user interface devices, UID **114**, and at least one user display **115** that is configured to display, for example, a view of the surgical site inside the patient **106**. In the example user console **102**, the remote operator **109** is sitting in the seat **110** and viewing the user display **115** while manipulating a foot-operated control **113** and a handheld UID **114** in order to remotely control the surgical robotic arms **104** and the surgical tools **107** (that are mounted on the distal ends of the surgical arms).

(16) In some variations, the bedside operator **108** may also operate the surgical robotic system **100** in an “over the bed” mode, in which the bedside operator **108** (user) is now at a side of the patient **106** and is simultaneously manipulating i) a robotically-driven tool (having an end effector) that is attached to the surgical robotic arm **104**, e.g., with a handheld UID **114** held in one hand, and ii) a manual laparoscopic tool. For example, the bedside operator's left hand may be manipulating the handheld UID to control a surgical robotic component, while the bedside operator's right hand may be manipulating a manual laparoscopic tool. Thus, in these variations, the bedside operator **108** may perform both robotic-assisted minimally invasive surgery and manual laparoscopic surgery on the patient **106**.

(17) During an example procedure (surgery), the patient **106** is prepped and draped in a sterile fashion to achieve anesthesia. Initial access to the surgical site may be performed manually while the arms of the surgical robotic system **100** are in a stowed configuration or withdrawn configuration (to facilitate access to the surgical site.) Once access is completed, initial positioning or preparation of the surgical robotic system **100** including its surgical robotic arms **104** may be performed. Next, the surgery proceeds with the remote operator **109** at the user console **102** utilizing the foot-operated controls **113** and the UIDs **114** to manipulate the various end effectors and perhaps an imaging system to perform the surgery. Manual assistance may also be provided at the procedure bed or table, by sterile-gowned bedside personnel, e.g., the bedside operator **108** who may perform tasks such as retracting tissues, performing manual repositioning, and tool exchange upon one or more of the surgical robotic arms **104**. Non-sterile personnel may also be present to assist the remote operator **109** at the user console **102**. When the procedure or surgery is completed, the surgical robotic system **100** and the user console **102** may be configured or set in a state to facilitate post-operative procedures such as cleaning or sterilization and healthcare record entry or

printout via the user console **102**.

(18) In one embodiment, the remote operator **109** holds and moves the UID **114** to provide an input command to move a robot arm actuator **117** in the surgical robotic system **100**. The UID **114** may be communicatively coupled to the rest of the surgical robotic system **100**, e.g., via a console computer system **116**. The UID **114** can generate spatial state signals corresponding to movement of the UID **114**, e.g. position and orientation of the handheld housing of the UID, and the spatial state signals may be input signals to control a motion of the robot arm actuator **117**. The surgical robotic system **100** may use control signals derived from the spatial state signals, to control proportional motion of the actuator **117**. In one embodiment, a console processor of the console computer system **116** receives the spatial state signals and generates the corresponding control signals. Based on these control signals, which control how the actuator **117** is energized to move a segment of the surgical robotic arm **104**, the movement of a corresponding surgical tool that is attached to the arm may mimic the movement of the UID **114**. Similarly, interaction between the remote operator **109** and the UID **114** can generate for example a grip control signal that causes a jaw of a grasper of the surgical tool **107** to close and grip the tissue of patient **106**.

(19) Surgical robotic system **100** may include several UIDs **114**, where respective control signals are generated for each UID that control the actuators and the surgical tool (end effector) of a respective surgical robotic arm **104**. For example, the remote operator **109** may move a first UID **114** to control the motion of an actuator **117** that is in a left robotic arm, where the actuator responds by moving linkages, gears, etc., in that surgical robotic arm **104**. Similarly, movement of a second UID **114** by the remote operator **109** controls the motion of another actuator **117**, which in turn moves other linkages, gears, etc., of the surgical robotic system **100**. The surgical robotic system **100** may include a right surgical robotic arm **104** that is secured to the bed or table to the right side of the patient, and a left surgical robotic arm **104** that is at the left side of the patient. An actuator **117** may include one or more motors that are controlled so that they drive the rotation of a joint of the surgical robotic arm **104**, to for example change, relative to the patient, an orientation of an endoscope or a grasper of the surgical tool **107** that is attached to that arm. Motion of several actuators **117** in the same surgical robotic arm **104** can be controlled by the spatial state signals generated from a particular UID **114**. The UIDs **114** can also control motion of respective surgical tool graspers. For example, each UID **114** can generate a respective grip signal to control motion of an actuator, e.g., a linear actuator, that opens or closes jaws of the grasper at a distal end of surgical tool **107** to grip tissue within patient **106**.

(20) In some aspects, the communication between the surgical platform **105** and the user console **102** may be through a control tower **103**, which may translate user commands that are received from the user console **102** (and more particularly from the console computer system **116**) into robotic control commands that are transmitted to the surgical robotic arms **104** on the surgical platform **105**. The control tower **103** may also transmit status and feedback from the surgical platform **105** back to the user console **102**. The communication connections between the surgical platform **105**, the user console **102**, and the control tower **103** may be via wired and/or wireless links, using any suitable ones of a variety of data communication protocols. Any wired connections may be optionally built into the floor and/or walls or ceiling of the operating room. The surgical robotic system **100** may provide video output to one or more displays, including displays within the operating room as well as remote displays that are accessible via the Internet or other networks. The video output or feed may also be encrypted to ensure privacy and all or portions of the video output may be saved to a server or electronic healthcare record system.

(21) FIG. 2 is an illustration of a subsystem or a part of the surgical robotic system **100**, for detecting engagement of a robotic surgical tool **240** to a tool drive **230** of a surgical robotic arm **220**. The surgical robotic arm **220** may be one of the surgical robotic arms **104** of surgical robotic system **100** illustrated and discussed with respect to FIG. 1. The control unit **210** may be part of for example the control tower in FIG. 1. As discussed in more detail herein, the engagement may be

detected by control unit **210** based on one or more motor operating parameters of one or more actuators (e.g., actuator **238-j**) in the tool drive **230**.

(22) There is a tool drive **230** to which different surgical tools (e.g., surgical tool **240**, as well as other detachable surgical tools—not shown) may be selectively attached (one at a time.) This may be done by for example a human user holding the housing of the surgical tool **240** in her hand and moving the latter in the direction of arrow **280** shown until the outside surface of the surgical tool **240** in which there are one or more tool disks (e.g., tool disk **244-i**) comes into contact with the outside surface of the tool drive **230** in which there are one or more drive disks (e.g., drive disk **234-j**). In the example shown, the tool drive **230** is a segment of the surgical robotic arm **220** at a distal end portion of the surgical robotic arm **220**. A proximal end portion of the arm **220** is secured to a surgical robotic platform, such as a surgical table that is not shown in FIG. **2** but an example of which may be seen in FIG. **1** described above.

(23) Control unit **210** is responsible for controlling motion of the various motorized joints in the surgical robotic arm **220** (including the drive disks **234**) through which operation of end effector **246** (its position and orientation as well as its surgical function) which mimics that of a user input device is achieved. This is achieved via a mechanical transmission in the surgical tool **240**, when the surgical tool **240** has been engaged to transfer force or torque from the tool drive **230**. The control unit **210** may be implemented as a programmed processor, for example as part of the control tower **103** of FIG. **1**. It may respond to one or more user commands received via a local or remote user input (e.g., joystick, touch control, wearable device, or other user input device communicating via console computer system **116**.) Alternatively, the control unit **210** may respond to one or more autonomous commands or controls (e.g., received from a trained surgical machine learning model that is being executed by the control unit **210** or by the console computer system **116**), or a combination thereof. The commands dictate the movement of robotic arm **220** and operation of its attached end effector **246**.

(24) The end effector **246** may be any surgical instrument, such as jaws, a cutting tool, an endoscope, spreader, implant tool, etc. Different surgical tools each having different end effectors can be selectively attached (one at a time) to robotic arm **220** for use during a surgical or other medical procedure. The end effector **246** depicted in the example of FIG. **2** is jaws located at a distal end of the surgical tool **240** and that may be retracted into, or extend out of, a cannula as shown (e.g., a thin tube that may be inserted into a patient undergoing a surgical procedure).

(25) The robotic arm **220** includes a tool drive **230**, in which there are one or more actuators, such as actuator **238-j**. Each actuator may be a linear or rotary actuator that has one or more respective electric motors (e.g., a brushless permanent magnet dc motor) whose drive shaft may be coupled to a respective drive disk **234-j** through a transmission (e.g., a gear train that achieves a given gear reduction ratio—not shown). The tool drive **230** includes one or more drive disks **234** that may be arranged on a planar or flat surface of the tool drive **230**, wherein the figure shows several such drive disks that are arranged on the same plane of the flat surface. Each drive disk (e.g., drive disk **234-j**) is exposed on the outside surface of the tool drive **230** and is designed to mechanically engage (e.g., to securely fasten via snap, friction, or other mating features) a mating tool disk **244-j** of the surgical tool **240**, to enable direct torque transfer between the two. This may take place once for example a planar or flat surface of the surgical tool **240** and corresponding or mating planar or flat surface of the tool drive **230** are brought in contact with one another.

(26) Furthermore, a motor driver circuit (not shown but that may for example be installed in the tool drive **230** or elsewhere in the surgical robotic arm **220**) is electrically coupled to the input drive terminals of a constituent motor of one or more of the actuators **238**. The motor driver circuit manipulates the electrical power drawn by the motor in order to regulate for example the speed of the motor or its torque, in accordance with a motor driver circuit input, which can be set or controlled by control unit **210**, which results in the powered rotation of the associated drive disk (e.g., drive disk **234-j**).

(27) When the mating drive disk **234-j** is mechanically engaged to a respective tool disk **244-j**, the powered rotation of the drive disk **234-j** causes the tool disk **244-j** to rotate, e.g., the two disks may rotate as one, thereby imparting motion on, for example, linkages, gears, cables, chains, or other transmission means within the surgical tool **240** for controlling the movement and operation of the end effector **246** which may be mechanically coupled to the transmission means.

(28) Different surgical tools may have different numbers of tool disks based on the types of movements and the number of degrees of freedom in which the movements are performed by their end effectors, such as rotation, articulation, opening, closing, extension, retraction, applying pressure, etc.

(29) Furthermore, within the surgical tool **240**, more than one tool disk **244** may contribute to a single motion of the end effector **246** to achieve goals such as load sharing by two or more motors that are driving the mating drive disks **234**, respectively.

(30) In another aspect, within the tool drive **230**, there may be two or more motors whose drive shafts are coupled (via a transmission) to rotate the same output shaft (or drive disk **234**), to share a load.

(31) In yet another aspect, within the surgical tool **240**, there may be a transmission which translates torque from two drive disks **234** (via respective tool disks **244**) for performing complimentary actions in the same degree of freedom, e.g., a first drive disk **234-i** rotates a drum within the housing of the surgical tool **240** to take in one end of a cable, and a second drive disk **234-j** rotates another drum within the housing of the surgical tool **240** to take in the other end of the cable. As another example, the extension and the shortening of an end effector along a single axis may be achieved using two drive disks **234-i**, **234-j**, one to perform the extension and another to perform the retraction, for example via different cables. This is in contrast to an effector that also moves in one degree of freedom (e.g., extension and shortening longitudinally along a single axis of movement) but that only needs a single tool disk to control its full range of movement. As another example, an effector that moves in multiple degrees of freedom (e.g., such as a wristed movement, movement along multiple axes, activation of an energy emitter in addition to end effector movement, etc.) may necessitate the use of several tool disks (each being engaged to a respective drive disk). In another type of surgical tool **240**, a single tool disk **244** is sufficient to perform both extension and retraction motions, via direct input (e.g., gears). As another example, in the case of the end effector **246** being jaws, two or more tool disks **244** may cooperatively control the motion of the jaws, for load sharing, as discussed in greater detail herein.

(32) In some embodiments, when surgical tool **240** is first attached to or installed on tool drive **230** such that the tool disks are brought substantially into coplanar and coaxial alignment with corresponding drive disks (though the tool and drive disks are perhaps not yet successfully engaged), control unit **210** initially detects the type of the surgical tool **240**. In one embodiment, surgical tool **240** has an information storage unit **242**, such as a solid state memory, RFID tag, bar code (including two-dimensional or matrix barcodes), etc., that identifies its tool or end effector information, such as one or more of identification of tool or end effector type, unique tool or end effector ID, number of tool disks used, location of those tool disks being used (e.g., from a total of six possible tool disks **244**), type of transmission for the tool disks (e.g., direct drive, cable driven, etc.), what motion or actuation a tool disk imparts on the end effector, one or more tool calibration values (e.g., a rotational position of the tool disk as determined during factor testing/assembly of the tool), whether motion of the end effector is constrained by a maximum or minimum movement, as well as other tool attributes. In one embodiment, the information storage unit **242** identifies minimal information, such as a tool ID, which control unit **210** may use to perform a lookup of the various tool attributes.

(33) The tool drive **230** may include a communication interface **232** (e.g., a memory writer, a near field communications, NFC, transceiver, RFID scanner, barcode reader, etc.) to read the information from the information storage unit **242** and pass the information to control unit **210**.

Furthermore, in some embodiments, there may be more than one information storage unit in surgical tool **240**, such as one information storage unit associated with each tool disk **244**. In this embodiment, tool drive **230** may also include a corresponding sensor for each possible information storage unit that would be present in a given tool.

(34) Engagement

(35) After surgical tool **240** is attached with tool drive **230**, such that tool disks are brought into alignment and are superimposed on corresponding drive disks (although not necessarily mechanically engaged), and after the tool disk information is obtained, e.g., read by control unit **210**, the control unit **210** performs an engagement process to detect when all of the tool disks that are expected to be attached to respective drive disks are mechanically engaged with their respective drive disks (e.g., their mechanical engagement has been achieved, or the tool drive is now deemed engaged with the tool). That is, attaching the surgical tool **240** with the tool drive **230** does not necessarily ensure the proper mating needed for mechanical engagement of tool disks with corresponding drive disks (e.g., due to misalignment of mating features). The engagement process may include activating one or more motors of an actuator (e.g., actuator **238-j**) that drives a corresponding drive disk **234-j**. Then, based on one or more monitored motor operating parameters of the actuator **238-j**, while the latter is driving the drive disk **234-j**, the mechanical engagement of the tool disk **244-i** with a drive disk **234-j** can be detected, as discussed in greater detail below. This process may be repeated for every drive disk **234** (of the tool drive **230**) that is expected to be currently attached to a respective tool disk **244** (e.g., as determined based on the tool disk information obtained for the particular surgical tool **240** that is currently attached.)

(36) Upon detecting that a particular type of surgical tool **240** has been attached with the tool drive **230**, the control unit **210** activates one or more actuators (e.g., motors) of the tool drive **230** that have been previously associated with that type of surgical tool **240**. In some embodiments, each actuator that is associated with a corresponding drive disk **234** of surgical tool **240** may be activated simultaneously, serially, or a combination of simultaneous and serial activation. FIG. 3 illustrates an example of the surgical tool **240** that utilizes four tool disks, such as tool disk **244-i**, arranged in a coplanar fashion on a mating surface of its housing. Each tool disk contributes to at least a portion of the movement and/or activation of end effector **246**. Upon detecting the attachment of surgical tool **240** with tool drive **230** (e.g., joining of mating surfaces of the respective housings), control unit **210** (or its processor **312** while executing instructions stored in memory **314** as engagement control **316**) performs a process which determines that only the corresponding four drive disks, such as drive disk **234-j**, need to be turned (a corresponding actuator **238-j** needs to be activated-see FIG. 2) to perform the engagement process.

(37) Returning to FIG. 2, during operation of actuator **238-j**, after the detected attachment of surgical tool **240** with tool drive **230**, one or more sensors **236-j** measure one or more motor operating parameters of the actuator **238-j** as its motor is signaled to start to move. In one embodiment, the actuator **238-j** will turn in a direction that causes its attached (though not yet engaged) tool disk **244-i** to wind a cable in the transmission housing of the tool **240**, for cable driven surgical tools. As an example, see FIG. 4A in which motion **445** of the tool disk **244** avoids unwinding of cable **446**, and thereby holds the tool disk **244** in place or starts to turn the tool disk **244** in the direction of motion **445** (which winds the cable **446**.) This turning of the tool disk **244** continues until engagement is achieved as explained further below in connection with FIG. 4B and FIG. 4C.

(38) In another embodiment, the selected actuator is signaled to turn so as to cause its attached tool disk **244** to rotate so that the end effector **246** that is connected to the tool disk **244** moves towards a physical constraint (e.g. a jaw opens until it stops against a cannula wall, a maximum in a range of motion is achieved when bumping against a hard stop in a fully open position, etc.) In yet another embodiment, such as an endoscope embodiment where two actuators are sharing the load being rotation of an endoscope camera where there may be no hard stops against rotation of the

camera, the selected actuator rotates its attached tool disk **244-i** in a direction that opposes the motion of another tool disk **244-j** that is also rotatably coupled to the same output shaft in the transmission housing of the tool **240**. In that instance, as soon as one of the tool disks **244-i**, **244-j** engages, it will act as a physical constraint to the other tool disk. Other predetermined directions of movement may also be used consistent with the discussion herein.

(39) Furthermore, in some embodiments, the actuator's movement is ramped or increased gradually by the control unit **210** (e.g., the control unit **210** signals or commands the actuator to start to rotate at a slow speed at the beginning of movement and then progressively increase the speed, and then progressively decrease the speed at detection of engagement).

(40) In one embodiment, the calibration values stored in the information storage unit **242** of the surgical tool may be used to expedite tool engagement. For example, the calibration values can include a factory determined position (angle) of a particular tool disk **244-j**, recorded during product assembly or testing. The engagement process may need to have knowledge of a home position of a corresponding drive disk **234-j**. This knowledge may be obtained by the control unit **210** performing a tool driver calibration routine, in which it determines when a particular drive disk **234-j** has reached a home position (as the control unit actuates the drive disk **234-j**), such that position of that drive disk **234-j** is now known by the control unit **210**. Note that the control unit **210** may do so while only relying on output from a position sensor that is in the tool drive **230**, and the tool **240** itself may be passive in that it has no electronic sensors in it.

(41) Next, the control unit **210** may activate the corresponding actuator of the drive disk **234-j** so that the drive disk **234-j** turns at a high speed until a position variable of the drive disk **234-j** comes close to the factory determined position. When the drive disk satisfies a threshold distance relative to the factory determined position (e.g., a home position of the tool disk) which implies that mating features of the tool disk and drive disk are near alignment, the speed may be reduced so as to increase the likelihood that the mating features will engage one another upon their initial encounter. This process may work for both direct transmissions as well as for tool disks that utilize a cable to drive the effector (as in FIG. 4A.) For the latter, the calibration value may include a rotation count (e.g., a number of complete rotations of a motor drive shaft) that can be used to limit the continued turning of the drive disk **234-j** once engagement has been detected, to ensure that a maximum length of winding of the cable is not exceeded, or an angle of rotation of the engaged, tool disk **244-j** and drive disk **234-j** is not exceeded.

(42) In some embodiments, the motor operating parameters monitored by the control unit **210** (via sensors **236**) are interpreted to mean successful mechanical engagement of a tool disk with a drive disk. These can include measurements of torque applied by the actuator **238-j** as measured by a torque or force sensor, measurements of current supplied to a motor of the actuator **238-j** when attempting to drive the actuator to move at a certain velocity (e.g., where the sensor **236-j** may include a current sensing resistor in series with a motor input drive terminal), measurements of electrical impedance as seen into the input drive terminals of the motor of the actuator when attempting to drive the motor to move at a certain velocity (e.g., where the sensor **236-j** may also include a voltage sensing circuit to measure voltage of the motor input drive terminal), speed of the actuator **238-j** (e.g., where the sensor **236-j** may include a position encoder on an output shaft of the actuator **238-j** or on a drive shaft of the motor), as well as other parameters referred to here as motor operating parameters. While monitoring the one or more motor operating parameters of a particular actuator, when one or more of these parameters satisfies (e.g., meets or reaches) a predetermined, condition or threshold, the detection of such a situation can be interpreted by control unit **210** as a mechanical engagement event. Note that satisfying the predetermined condition may for example mean that the monitored operating parameter exhibits certain changes, as per the threshold, relative to an operating parameter of another motor that is part of the same actuator **238-j** or that is part of another actuator **238-i** which is being controlled by the control unit **210** simultaneously during the engagement detection process.

(43) In some embodiments, detection of certain motor operating parameters during operation of the actuator **238-j**, such as one or more of i) torque that satisfies (e.g., rises and reaches) a torque threshold, ii) motor current that satisfies (e.g., rises and reaches) a current threshold, iii) impedance that drops below an impedance threshold, iv) motor speed dropping below a motor velocity threshold, or a combination thereof, are used by control unit **210** to determine that mechanical engagement of tool disk **244-j** to drive disk **234-j** has occurred. The following are some examples of such a process.

(44) In one embodiment, where the tool disk **244-j** uses a cable **446** to control movement of its end effector **246**, the actuator **238-j** (that is driving the corresponding drive disk **234-j**) will move in the direction that winds the cable (here, the direction of motion **445** of which the control unit **210** may have knowledge, based on having previously identified the type of tool **240**). FIG. **4A** depicts such a tool disk **244** that has a pair of coupling features **447a**, **447b** on its disk face, which are depicted as hollow circles. Each coupling feature **447a**, **447b** may be a separate, cylindrical cavity formed in the disk face. The direction of motion **445** will wind the cable **446** around it, where in this initial condition the cable **446** has some slack as shown that will disappear as the cable **446** is being wound in the direction of motion **445**.

(45) A drive disk **234** is concentrically aligned with the tool disk **244**, as seen in FIG. **4B**. That is, FIG. **4B** illustrates drive disk **234** aligned and superimposed over the tool disk **244** such that their respective disk faces are brought into contact with one another. The drive disk **234** has a pair of coupling features **448a**, **448b** on its disk face, depicted as solid circles. Each coupling feature **448a**, **448b** may be a separate, cylindrical pin formed on the disk face. In this particular example, each of the coupling features **448** is sized to easily fit into either of the features **447** (once the two complementary features are aligned.) In FIG. **4B**, the features **447a**, **447b** and the features **448a**, **448b** are misaligned even though the faces of their respective tool and drive disks are in contact with one another. In other words, in FIG. **4B** one or more mating or complementary pairs of features, such as features **447a-448a**, or features **447b-448b**, are not yet mechanically engaged with each other. During such a condition, the drive disk **234** continues to be driven by its actuator **238**, to move (turn) in the direction of motion **445** until mechanical engagement is reached in the condition depicted in FIG. **4C**.

(46) As illustrated in FIG. **4C**, the drive disk **234** while turning has reached the point where both the coupling features **447a-448a** and the coupling features **447b-448b** have mechanically engaged each other as shown so that they now move as one (if the drive disk **234** continues to turn.) In the example here, each pin-cavity pair is now interlocked as shown in this figure. In addition, at this point the cable **446** is now taught after having been wound, and may thus serve to help hold the tool disk **244** in place (prevents its rotation) as drive disk **234** continues to turn in the direction of motion **445**. Further turning of the drive disk **234** in the condition of FIG. **4C** may increase tension in the cable **446** as the cable **446** pulls on its end effector **246** until a hard stop is reached which creates a physical constraint against further movement in the direction of motion **445** of the now-engaged drive disk **234**.

(47) The physical constraint on further turning of the drive disk **234** enables detection of the mechanical engagement event, by the control unit **210** making its measurement of motor operating parameters and comparison to one or more thresholds that may have been predetermined to be indicative of engagement. For example, the velocity/speed of the motor dropping below one or more threshold values indicates engagement because the motor is constrained from further movement in the winding direction. As another example, engagement occurs when the torque applied by the motor increases to a value greater than a freely moving motor and/or greater than the friction that results from tool disks and drive disks and/or attachment features rubbing or sliding against one another before engagement. Similarly, in other embodiments, the measured current and/or impedance approaches and may reach a maximum predetermined value when engagement occurs and as power is continued to be supplied to a motor in an attempt to continue movement of

the drive disk in the predetermined direction. When one or more of these thresholds are satisfied, control unit **210** can conclude that engagement has occurred between a tool disk and a drive disk. (48) Other forms of physical constraint can be used by control unit **210** for detecting successful drive disk and tool disk engagement. For example, a motion constraint, such as a mechanical limit of a range of motion imposed by a joint of the end effector (e.g., a joint that can only rotate about an axis from -45 to 45 degrees) or a physical barrier to movement (e.g., a cannula wall that impedes movement of the end effector), may also be utilized as the physical constraint/hard stop discussed above for either cable driven or non-cable driven tools.

(49) In some embodiments, the surgical tool **240** may not have a physical constraint/hard stop in at least one degree of freedom of movement from which motor operating parameters can be measured. For example, a tool disk **244-j** may be responsible for imparting unconstrained rotation of the end effector **246** element about an axis. Even with such unconstrained movement however, the control unit **210** may still detect engagement of the drive disk **234-j** to the tool disk **244-j** by detecting changes in one or more motor operating parameters during the engagement detection process. For example, a motor operating parameter pattern, such as repeating torque spikes caused by the features **447** rotating past the features **488** (and thus not engaging each other) is indicative of lack of engagement. Accordingly, the stopping or non-existence of that torque spike pattern (while the drive disk **234-j** continues to turn) means that the control unit **210** has detected tool disk to drive disk engagement.

(50) In some embodiments, a physical constraint may be created by the use of coordinating movement of multiple drive disks, and/or by letting a single drive disk engage before engaging a second drive disk. For example, consider the case where two or more tool disks (in the same housing of a surgical tool **240**) are connected by a transmission in the housing of the tool **240** to share a load (end effector **246**) when turning in the same direction, such as when a cutting or clamping tool may need to apply force beyond that which a single actuator **238-j** could supply. In such an embodiment, two or more actuators that are turning in the same direction (their respective drive disks are turning in the same direction) are driving the same output shaft that is inside the surgical tool **240** (due to the transmission in the surgical tool **240** that is connected to the corresponding tool disks.) Now, if the two actuators are signaled to move in opposing directions, then as soon as one of the drive disks engages its corresponding tool disk, this becomes a physical constraint to the other drive disk (when the other drive disk has engaged its corresponding tool disk.) When one of the two or more actuators engages (its drive disk engages its corresponding tool disk), the control unit **210** creates a constraint for the other actuator by signaling the engaged actuator to, for example, enter a position hold state. That is, a first actuator **238-j** will be commanded by the control unit **210** to hold its position while the other, non-engaged actuator **238-i** continues to be signaled to drive and thus turn or move (toward engagement between its drive disk **234-i** and tool disk **244-i**.) In this embodiment, one or both of the actuators' motor operating parameters can be monitored to detect engagement between a tool disk and drive disk pair. Furthermore, if a hard stop does exist (the control unit **210** expects or knows that this particular tool **240** has a hard stop), then the actuator of an engaged drive disk can be signaled to continue to drive or turn in the same direction until the hard stop is detected. The other actuator may continue to turn in the opposing direction and attempt to engage while the already engaged actuator holds its position at the hard stop.

(51) Returning to FIG. 2, and as discussed above, the tool identification performed by control unit **210** enables the latter to get knowledge of the characteristics of the end effector **246** of surgical tool **240**. For example, the control unit **210** may use that identification process to determine whether two (or more) tool disks in the tool **240** act in concert to impart end effector movement, whether one or more movements of the end effector are subject to hard stops or physical constraints, what are the ranges of movement of the end effector, what actuators will be used by the tool **240**, and factory defined calibration values such as a home position of a tool disk. Note that a calibration

value may encompass a range, e.g., 290 degrees $\pm$ 4 degrees. Based on such a calibration value, e.g., a home position of the tool disk **244-i** and based on the present position of the corresponding drive disk **234-i** (determined using a position encoder in the tool drive **230**), the control unit **210** can track the difference during the engagement process (as the actuator is signaled to turn.) So long as the difference is greater than a predetermined threshold, then the actuator is signaled to turn rapidly (fast rotation), and then in response to the difference becoming smaller than the threshold (implying that the drive disk is nearing the calibration value home position) the actuator is signaled to turn slowly (slow rotation). And that is expected to increase the chances of a reliable engagement being detected.

(52) In some embodiments, after an engagement is detected by control unit **210**, control unit **210** may take one or more additional actions with respect to the end effector **246** to confirm the engagement. For example, control unit **210** may subject the end effector to a predetermined set of one or more motions to test the engagement, such as signaling a drive disk to reverse direction thereby moving end effector in an opposite direction to what it was doing during the engagement process, moving the end effector to achieve an expected maximum degree of movement, etc. Such movements enable control unit **210** to for example reach a hard stop or reach a physical constraint again, which is detected as discussed herein based on one or more motor operating parameters, to confirm mechanical engagement between the tool disks and drive disks.

(53) Furthermore, in some embodiments, control unit **210** may utilize the hard stop or physical constraint to set a reference position of the end effector. For example, knowing that a hard stop is to occur when the end effector reaches 270° of rotation in a certain direction, control unit **210** can set calibration values for a position of the corresponding actuator or drive disk. Then, movement of the actuator or drive disk can be tracked based on number of rotations of the drive disk, motor shaft, gear ratio, drive disk/motor indexing, etc.

(54) Furthermore, in some embodiments, control unit **210** may signal actuation by one or more motors for a specified number of times, a specified number of rotations, or a combination thereof, when attempting to achieve engagement of tool disks with drive disks. When engagement is not achieved within a threshold amount of time, number of rotations, etc. control unit **210** may issue a warning to an operator of the surgical robotic system (e.g., an operator of system **100** of FIG. 1) to detach and then reattach surgical tool **240** to restart the engagement process.

(55) After mechanical engagement of drive disks with tool disks is detected by control unit **210**, an operator may command motions of one or more joints of surgical robotic arm **220**. As discussed above, the commands are received from or originate from one or more UIDs (e.g., UID **114**), as spatial state signals from the UIDs which are translated to corresponding control signals that the control unit **210** provides (e.g., a desired motor speed or current and direction of rotation) to energize one or more actuators of tool drive **230** which will change the pose, position or other state of the end effector. In one embodiment, where two or more actuators are cooperatively controlling the motion of the end effector, such as when two or more tool disks are to impart motion of the end effector in the same degree of freedom, control unit **210** further performs a cooperative control technique to ensure that the actuators operate in a complementary fashion when moving the end effector, share a load associated with movement of the end effector, do not fight one another in imparting such motion, maintain a balance between the actuators so that one actuator does not continually perform more or less work than the other actuators, etc. For example, when two or more actuators are used to control the opening, closing, and application of grip force of jaws of the end effector **246**, control unit utilizes a multi-actuator operation control technique that identifies a first of the two or more actuators as a master actuator, and the remaining one or more actuators as slave actuators. Then, a position command that has been provided to signal the master actuator to move the end effector **246** to a commanded position, is also provided to signal the slave actuator to move the end effector **246** to the same commanded position. For instance, if the master actuator and the slave actuator are replicates, then if the master actuator receives a certain polarity (direction

of rotation of its motor) and a certain motor current value to satisfy a given end effector position command, the same polarity and current value may also be supplied to each of the slave actuators. In some embodiments however, there may be some compensation for how motion of the actuators complement each other, for example, reversing polarity for the slaves when rotation directions of master and slave actuators are different, adjusting gain (e.g., of the commanded motor current) when attributes of the motors are different, etc., as discussed in greater detail herein.

(56) FIG. 3 is a block diagram showing an example of the surgical tool **240**, tool drive **230**, and control unit **210**. The surgical tool **240** may be attached with the tool drive **230** by bringing complementary or mating surfaces of their respective housings in contact within one another. The attaching may also include fastening the housings with one another. Furthermore, one or more sensors (not shown) of the tool drive **230** may be used by the control unit **210** to detect the attaching, including reading data from the surgical tool **240** that identifies the surgical tool **240**, indicates which tool disks (e.g., tool disk **244-j**) are used to control movements of the end effector **246**, includes calibration values, indicates whether the tool has hard stops, or indicates which tool disks contribute to movement of or are connected by a transmission to other tool disks in the surgical tool **240**. The data may be transferred to the control unit **210** via a communication link (e.g., a wired or wireless link) established between a communications interface **318** of the control unit **210** and sensor readout circuitry (not shown) in the tool drive **230**. The data may then be stored in memory **314** as part of an engagement control program (engagement control **316**) and may be associated with that particular surgical tool **240** so long as the latter remains attached to the tool drive **230**.

(57) The control unit **210** including its programmed processor **312** may be integrated into the surgical robotic system **100** (FIG. 1) for example as a shared microprocessor and program memory within the control tower **103**. Alternatively, the control unit **210** may be implemented in a remote computer such as in a different room than the operating room, or in a different building than the operating arena shown in FIG. 1. Furthermore, control unit **210** may also include, although not illustrated, user interface hardware (e.g., keyboard, touch-screen, microphones, speakers) that may enable manual control of the robotic arm and its attached tool **240**, a power device (e.g., a battery), as well as other components typically associated with electronic devices for controlling surgical robotic systems.

(58) Memory **314** is coupled to one or more processors **312** (generically referred to here as “a processor” for simplicity) to store instructions for execution by the processors **312**. In some embodiments, the memory is non-transitory, and may store one or more program modules, including a tool control **320** and an engagement control **316**, whose instructions configure the processor **312** to perform the engagement processes described herein. In other words, the processor **312** may operate under the control of a program, routine, or the execution of instructions stored in the memory **314** as part of the tool control **320** and engagement control **316** to execute methods or processes in accordance with the aspects and features described herein.

(59) In response to detecting the attaching of the surgical tool **240** with the tool drive **230**, engagement control **316** performs (or rather configures the processor **312** to perform) a process for detecting the mechanical engagement of tool disks with corresponding drive disks (which are actuator driven), such as engagement of tool disk **344-i** with corresponding drive disk **334-i**. The engagement control **316** may signal (through the tool control **320**) that one or more of the actuators of tool drive **230** impart motion of their respective drive disks. In some embodiments, these instructions or signals include instructions to energize, activate or otherwise provide power to a motor so that the motor can produce or apply a specific amount of torque, cause the drive disk to rotate at a specific speed and direction, by applying a certain voltage command, current command, etc. Furthermore, the motion of each drive disk can be controlled to start rapidly initially during the engagement detection process, and then ramp down slowly once engagement is near, or proximity to alignment of mating features is detected or a predetermined time limit is reached without

detecting engagement. For instance, based on the relative position of a drive disk to a tool disk (which may be based on a known calibration value), the actuator speed is ramped down to a predetermined speed (e.g., until the drive disk is within a threshold distance of where the mating features become aligned).

(60) The engagement control **316** monitors one or more motor operating parameters of the motors of actuators of the tool drive **230**. As discussed herein, the motor operating parameters can include torque imparted by a motor, voltage supplied to a motor, impedance as seen on the input drive terminals of a motor when attempting to drive the motor to move at a certain velocity, motor speed, as well as other motor operating parameters. One or more of these parameters may be monitored by comparing them to thresholds, so that when the thresholds are reached then a mechanical engagement event is deemed to have occurred (between, for example, tool disk **344-i** and drive disk **334-i**.) As discussed herein, mechanical engagement is expected to be detected when corresponding mating features of a tool disk and a drive disk align and fasten with one another to that rotation of the drive disk will cause for example both immediate and proportional rotation of the mechanically engaged tool disk (as one with the drive disk.) Such engagement is expected to be detected when one or more of the motor operating parameters satisfies a threshold (e.g., reaching or exceeding a threshold indicative of a hard stop being reached, a maximum torque, voltage, or impedance value, a torque, voltage, or impedance value greater than what would be needed to overcome friction that initially appears when the tool **240** is first attached to the tool drive **230**. The engagement control **316** thus infers or deduces that tool disk **344-i** and drive disk **334-j** have engaged with one another (e.g., fastening of respective disk mating features with one another).

(61) Note that engagement control **316** need not monitor sensor readings for all of the motor parameters that are available from the tool drive **230**. Instead engagement control **316** could monitor only one or more characteristics of interest based on, for example, whether surgical tool **240** is subject to any hard stops or physical movement constraints, whether one or more tool disks operate in concert (cooperate with each other) to impart movement on surgical tool **240**, whether tool disks impart movement on surgical tool **240** via cables or directly (e.g., through a gear box), or a combination thereof, in order to determine when a threshold associated with engagement is satisfied.

(62) In some embodiments, engagement control **316** monitors patterns of motor operating parameters, such as patterns of torque, voltage, motor speed, impedance, etc. that are the result of drive disk **234-j** rotating over tool disk **244-j** but without mechanical engagement. That is, a certain amount of torque, force, voltage, etc. could be measured, which is greater than what is exhibited by a free moving drive disk (where the surgical tool **240** is not attached to the tool drive **230**) and less than a mechanically engaged drive disk (when for example mating features of the tool and drive disks pass each other while the tool and tool drive housings are in contact, but without engaging.) When this motor parameter pattern changes as detected, such as due to a hard stop or physical motion constraint being encountered, engagement control **316** is said to have detected mechanical engagement. Monitoring and interpreting pattern based motor operating pattern also enables engagement control **316** to detect engagement (between tool disk **344-j** and drive disk **334-j**) even when no hard stop or motion constraint is available, or without having to drive a drive disk to a tool's hard stop or other motion constraint.

(63) In some embodiments, when engagement control **316** detects mechanical engagement of tool disks with drive disks, it may also initiate a verification process or engagement check in which the actuators of tool drive **230** are signaled to undergo a predetermined set of one or more motions, to verify the detected engagement. For example, the actuators may be instructed to cause their respective drive disks to rotate in a direction opposite to the direction of engagement (the latter being the direction in which the drive disks were rotating when engagement was initially detected, e.g., in the direction of motion **445** seen in FIG. 4A). Then, the drive disks may be rotated back in the direction of engagement until a second engagement is detected (e.g., when a particular motor

parameter reaches a threshold that is consistent with the tool disk reaching a hard stop or a motion constraint, when a particular motor parameter reaches a threshold that is consistent with resistance against the rotating tool disk that is not caused by friction alone (friction between the tool disk and a corresponding drive disk), or a combination thereof.

(64) Engagement control **316**, based on having detected engagement of tool disks to drive disks, or based on a countdown timer having expired without detecting engagement, generates a notification for an operator of the surgical robotic system. The notification may either indicate that engagement has occurred so that the surgical tool **240** is ready for use, or that engagement has not occurred and so the surgical tool **240** should be reattached.

(65) FIG. 5A is a flow diagram illustrating a process **500** for engaging a surgical tool with a tool drive of a surgical robotic system, in accordance with an embodiment of the disclosure. The process **500** may be performed by a programmed processor (also referred to here as processing logic), configured according to software stored in memory (e.g., the processor **312** and the memory **314** of FIG. 3, where the processor **312** is configured according to the instructions of the tool control **320** and the engagement control **316**.)

(66) Referring to FIG. 5A, processing logic begins by activating an actuator of the tool drive to rotate a drive disk of the tool drive (processing block **502**). For example, processing logic may activate a linear or rotary actuator of a tool drive (e.g., tool drive **230**) to turn or rotate the drive disk (e.g., drive disk **234-j**). Furthermore, as discussed herein, when mechanically engaged, the rotation of a drive disk (e.g., disk **234-j**) will cause immediate or direct rotation of a corresponding tool disk (e.g., disk **244-j**) of a surgical tool (e.g. surgical tool **240**).

(67) Processing logic monitors one or more motor operating parameters of the actuator that is causing the rotation of the drive disk while activating the motor (processing block **504**). In some embodiments, the operating parameters of the motor being monitored can include torque, motor current, motor velocity, or a combination thereof.

(68) Based on the one or more monitored motor operating parameters, processing logic detects when the drive disk becomes mechanically engaged with the tool disk (processing block **506**). In one embodiment, the detection occurs when or in response to at least one of the one or more motor operating parameters being monitored satisfying a corresponding condition or threshold. For example, the condition can be associated with a value of a motor operating parameter that occurs in response to the motor reaching a physical constraint against further rotation of the tool disk (e.g., reaching a mechanical limit of a range of motion when a physical barrier to the movement is encountered, a maximum degree of movement of the end effector of the tool is reached, opposition with another activated motor actuator occurs, etc.). As another example, the condition may represent a motor operating parameter exhibited when there is friction due to the drive disk contacting and sliding against the tool disk during rotation but without the mechanical latching or fastening of the drive disk to the tool disk. In embodiments discussed herein, when mechanical engagement of the drive disk with the tool disk is detected, one or more additional actions, such as generating system or operator notifications, initiating one or more engagement verification operations, storing reference values, etc. may be performed by processing logic.

(69) FIG. 5B is a flow chart illustrating a process **550** for detecting engagement of a tool disk with a drive disk based on one or more motor operating parameters of an actuator that is driving the drive disk. The process **550** is performed by processing logic that may comprise any combination of hardwired circuitry and programmed processor, where for example the process **550** may be performed by the processor **312** programmed in accordance with the tool control **320** and engagement control **316** described above. The process may begin with detecting that a detachable surgical tool has been attached to a tool drive of a robotic arm of a surgical robotic system (processing block **552**). The attachment may be detected based on sensors of the tool drive coming within wireless detection range or being conductively connected with an information storage unit in the detachable surgical tool. As discussed herein, the information storage unit may include a tool

identifier, and may also include additional tool attributes, such as which of several available tool disks in the tool housing are actually connected by a transmission in the housing to the end effector in the detachable surgical tool, what type of transmission in the tool controls movement of the end effector (e.g., cable driven, direct drive, etc.), what direction of movement or rotation is allowed, if there are any ranges of such movement or rotation, calibration values (e.g., cable lengths, current cable length, maximum winding, rotational position of tool disks or a home position of a tool disk, etc.), as well as other tool attributes discussed herein.

(70) At least one motor of the tool drive is then activated causing the at least one motor to rotate an associated drive disk corresponding to a tool disk (that is connected by a transmission in the surgical tool to control motion of the end effector (processing block **554**). In one embodiment, a current to be supplied to the motor, a torque to be achieved by the motor, or a direction of movement is signaled to the tool control **320**, so that the motor will cause the drive disk to rotate at a predetermined velocity in a predetermined direction. In other words, processing logic causes a signal to be sent to a motor driver circuit, commanding the motor driver circuit to apply power to or energize the motor. In some embodiments, the predetermined speed is set based on a determination, at the time of the detected attachment of the tool drive with the detachable surgical tool, of for example, the type of tool, tool drive transmission type (e.g., cable driven, direct drive, etc.), type of restraint that will be encountered (e.g., a hard stop, a physical constraint, opposing motion constraint), or a combination of such factors. One or more motor operating parameters of the at least one motor of the tool drive are then monitored (processing block **556**). The monitored motor operating parameters may correspond with those being controlled by processing logic to cause motion of the motor (e.g., torque, speed.)

(71) Returning to FIG. 5B, the processing logic repeatedly checks to see whether or not an engagement condition has been met, e.g., a monitored motor operating parameter has reached a threshold (processing block **558**.) If so, then a mechanical engagement event is flagged, signifying that the drive disk has mechanically engaged with its corresponding tool disk (processing block **564**). As discussed herein, the threshold is indicative of a condition associated with the mechanical engagement of the drive disks. For example, mechanical engagement is expected when the torque, current, or impedance associated with the motor at a predetermined velocity or speed exceeds their values that are associated the motor encountering tool drive to tool disk friction alone. The threshold may be a value which is greater than the torque, voltage, impedance, etc. needed to overcome such friction. As another example, movement of the end effector may be subject to a physical constraint, such as a maximum range of motion of a joint, a hard stop (e.g., as imposed by a cannula wall), an opposing motion of another drive disk, as well as other physical constraints. In that case, the speed and direction of movement of the motor are selected to advance the end effector or tool disk towards the physical constraint. Then, when the physical constraint is reached, the monitored torque, current, impedance will spike to a maximum, while the speed will drop to zero. In this example, one threshold may refer to torque or motor current that is set near its maximum value, and another threshold may refer to velocity that is set lower than a nominal speed of the motor during rotation of the drive disk, e.g., substantially zero. Thus, the two thresholds act as a check against one another to make the detection of engagement more robust.

(72) In response to the detected engagement of the drive disk with the corresponding tool disk, the motion of the drive disk is stopped (processing block **566**). In one embodiment, when the motion is stopped, one or more reference values associated with that position or state of the end effector may be stored for later reference and use. For example, where a physical constraint was used to detect the engagement, an index value of the motor, a rotation count, etc. at that moment can be stored, and used later for re-locating the end effector at or near the physical constraint. The physical constraint may be, e.g., maximum cable length, cannula wall, maximum of a range of motion, etc. Furthermore, to prevent excessive tensioning of a cable in the case of a cable driven tool, the motion of the drive disk may be stopped, or the motor deactivated, simultaneously or nearly

simultaneously in response to the detection performed at block **558**.

(73) Once engagement of all of the relevant drive disks (those that correspond to in-use tool disks of the particular surgical tool) has been detected in processing block **567** (where the process described above in blocks **554-556-558-564-566** may have been performed for each respective drive disk) then a notification of tool engagement is then generated (processing block **568**). The notification may be a visual notification (e.g., a graphical user interface notification), an audible notification (e.g., a tone, sound, etc.), sensory (e.g., a haptic notification), or a combination of such generated by user interface hardware of the surgical robotic system.

(74) Returning briefly to processing block **558**, when the engagement condition is not met (e.g., a monitored motor operating parameter does not satisfy a threshold such that mechanical engagement of a tool disk and a corresponding drive disk has not occurred), a determination of whether a time or rotation limit has been reached is performed (processing block **560**). A failure to engage may be due to a broken cable, a tool disk and drive disk not positioned close enough to each other to allow for engagement, etc.) The time limit may be a predetermined maximum time interval (countdown timer value) in which a drive disk is allowed to rotate without detecting mechanical engagement with a tool disk. Similarly, the rotation limit may be a number of motor rotations necessary to impart one or more full rotations of its respective drive disk. For example, if the rotation limit is associated with one full rotation of the drive disk, it is assumed that engagement should occur within a single revolution of a drive disk. If the time limit, rotation limit, or some combination of limits are not reached (processing block **560**), the monitoring of the one or more motor operating parameter values continues (return to processing block **556**.) However, if the one or more limits are reached (processing block **560**), a notification, similar to the notification of processing block **568**, that an error has occurred and tool engagement has failed is generated (processing block **562**). In this case, an operator of the surgical system may be instructed to detach the surgical tool from the tool drive, and then re-attach them to restart the engagement process of FIG. 5B.

(75) Turning now to FIG. 5C, this is a diagram of a process performed by a control unit for engaging a surgical robotic tool with a tool drive, as part of a surgical robotic system. The system includes the surgical tool **240** that is depicted in FIG. 2, having one or more tool disks at a proximal end and an end effector at a distal end thereof as shown. A tool drive (e.g., tool drive **230**) is mounted at a distal end of a surgical robotic arm **220** as shown, where the tool drive **230** has one or more drive disks **234** each driven by a rotary motor within a housing of the tool drive. Each drive disk **234** is to be attached to a tool disk **244** of the surgical tool **240** to impart motion to the end effector **246**.

(76) Staying with FIG. 5C, the process for engaging a tool disk with a drive disk is performed by a control unit, and in particular by one or more processors of the control unit that are configured to (or programmed to) do so. Operation may begin with detecting that the surgical tool is attached to the tool drive (block **582**); this may be done by the processor wirelessly or via a wired connection reading an identification or other attributes of the tool, which has been brought into contact with the tool drive such that a tool disk comes into contact with a corresponding or respective drive disk. The control unit may then actuate each drive disk through the rotary motor (block **584**), and detect, during the actuation, that a drive disk is engaged to a respective tool disk (block **586**); the drive disk is said to be engaged to the tool disk when a pair of coupling features of the drive disk and the tool disk (there may be more than one pair, e.g., two as shown in FIG. 4A-FIG. 4C, or more) become interlocked. To detect the engagement, the control unit recognizes sensed changes in motor status (status of the actuator, or its motor operating parameters), including that a velocity of the rotary motor drops below a predetermined velocity threshold and a torque of the rotary motor rises above a predetermined torque threshold. The velocity threshold may correspond to a velocity at which the motor is substantially stopped, wherein the torque threshold is a value between i) a minimum torque for the motor to overcome the friction between the drive disk and the tool disk before engagement and ii) a maximum torque for the motor to produce. The changes in motor

status may be caused by at least one of: the end effector reaching a joint limit, an external force on the end effector, a motion constraint due to another motor of the tool drive being activated, and a combination thereof. Depending on the particular surgical tool, there may be more than one tool disk that is in use for operating the end effector. In that case, in block **587**, the engagement process described above is also performed for each additional tool disk (which has a corresponding drive disk in the tool drive.) The process then continues with block **588** where the control unit signals for example a user interface subsystem of the surgical robotic system to report engagement of the surgical tool but only if all of the tool disks that are in use for the particular tool have been detected as engaged with their respective drive disks.

(77) In one embodiment, a feedback loop may be used to monitor one or more motor operating parameters and detect when a threshold has been reached. FIG. **6** depicts a block diagram of a feedback loop used to control the velocity of a motor of a tool drive using velocity feedback. The feedback loop may be implemented in hardware, firmware, software, or a combination thereof. A velocity command is received by a controller **602**. Controller **602** may be, for example, a proportional-integral-derivative controller (e.g., PID controller **702** of FIG. **7**) providing a loop/feedback mechanism to provide an appropriate motor current (e.g., the controller output, ctrlr out) to drive motor/actuator **604** at the velocity and in the direction of the velocity command (also referred to as velocity setpoint.) For example, the direction may be in a winding direction of a cable driven tool disk **244-j**. As another example, the direction may be a direction that will oppose the motion of another motor whose attached tool disk **244-i** should be cooperating with the tool disk **244-j**. As yet another example, the direction may be a direction that causes the motor to advance the end effector towards a hard stop, such as a physical barrier or a mechanical limit of range of motion. In one embodiment, the controller **602** may use various values, such as desired torque to achieve the velocity, current to achieve the velocity, impedance indicative of a velocity, etc. as a measure for generating the controller's current output to motor/actuator **604**.

(78) A sensor, such as a torque sensor, velocity sensor, or a combination of sensors, measures the actual velocity of the motor/actuator. The actual velocity is then provided as feedback back to controller **602**, which may calculate an error based on a disparity between the actual velocity of the motor and the commanded velocity. The controller **602** responds to the disparity by adjusting its controller output, e.g., a motor current command to the motor/actuator **604**, a torque to be achieved by the motor/actuator **604**, an impedance value, etc. that will cause the motor/actuator **604** to move towards the velocity command or setpoint. In some embodiments, controller **602** may output a motor operating parameter, such as the torque, current, impedance, velocity, etc., calculated as a result of executing the feedback loop.

(79) In another embodiment, controller **602** can include a saturation block (not shown) to ensure that the controller output (e.g., a value that is controlling the motor current) does not exceed a threshold, e.g. a current threshold, torque threshold, impedance threshold. The value used by or input to the saturation block may be dual-purposed, namely also used as the motor operating parameter value supplied to the processor (for purposes of being monitored during the engagement process.)

(80) FIG. **7** depicts a block diagram of a feedback loop that includes a controller **702** for controlling the velocity of a motor of a tool drive. In one embodiment, controller **702** is a proportional-integral (PI) controller and may be used within controller **602**, and may be implemented in part as hardware, firmware, software, or a combination thereof.

(81) A velocity command (e.g., control variable setpoint) is received by controller **702**. Controller **702** provides a loop/feedback mechanism to adjust and provide an appropriate current (e.g., the controller output) to drive motor/actuator **704** at the velocity and in the direction of the velocity command/setpoint. For example, the direction may be in a winding direction of a cable driven surgical tool's tool disk. As another example, the direction may be a direction that will oppose the motion of another motor of the tool drive. As yet another example, the direction may be a direction

that causes the motor to advance the end effector towards a hard stop, such as a physical barrier or a mechanical limit of range of motion. In one embodiment, the controller **702** may use various values, such as desired torque to achieve the velocity, current to achieve the velocity, impedance indicative of a velocity, etc. as a measure for generating the controller's current output to motor/actuator **704**.

(82) Adjustments are made to the original velocity command, such as a proportional adjustment (e.g., block k.sub.p) to adjust the velocity proportional to an error (e.g., as determined by the feedback), as well as an integral adjustment (e.g., block k.sub.i) to adjust the velocity to account for past error integrated over time. The integral adjustment may further be adjusted using a restoring term generated by block k.sub.b which is in feedback loop for anti-windup, to further adjust the value of the integral adjustment. The integral adjusted value may be further adjusted by block 1/s (e.g., before being added to the proportional setpoint adjusted value. In one embodiment, a saturation block may be used in controller **702** as shown, to ensure that a value controlling the current supplied to the motor does not exceed a threshold, e.g. a current threshold, torque threshold, impedance threshold, etc. After the adjustments are carried out by the blocks discussed above, and the resulting current command value does not exceed the value set in the saturation block, the current command (e.g., the adjusted command, which has been corrected based on the feedback and PI adjustments) is fed into the current amplifier, so that the commanded current into the motor actuator **704** can be amplified by a factor. The factor may be fixed, based on properties of the motor/actuator **704**, based on feedback from the motor/actuator **704**, etc. The motor/actuator **704** is activated as per the amplified current, and feedback (e.g., speed, torque, velocity, etc. as determined by a sensor coupled with a motor) on the velocity response of the motor/actuator **704** is provided to the feedback loop implemented by controller **702** as shown.

(83) As discussed above, the value used by (or input to) the saturation block may be used as the motor operating parameter value supplied to the processor for purposes of monitoring during the engagement process, e.g., representing present motor current or present motor impedance.

(84) In another embodiment, the feedback may be used as a motor operating parameter value supplied to the processor for purposes of monitoring during the engagement process. Other variables computed in the controller **702** (e.g., adjusted and non-adjusted) may be used as a monitored motor operating parameter.

(85) The above description of illustrated embodiments of the invention, including what is described below in the Abstract, is not intended to be exhaustive or to limit the invention to the precise forms disclosed. While specific embodiments of, and examples for, the invention are described herein for illustrative purposes, various modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize. For example, although FIG. 4A-FIG. 4C depict a surgical tool **240** that has a cable-driven transmission connecting the tool disk **244** to the end effectors (not shown), the engagement process described above is also applicable to other types of surgical tools having different transmissions (not necessarily cable-driven.) These modifications can be made to the invention in light of the above detailed description. The terms used in the following claims should not be construed to limit the invention to the specific embodiments disclosed in the specification. Rather, the scope of the invention is to be determined entirely by the following claims, which are to be construed in accordance with established doctrines of claim interpretation.

Claims

1. A method for engaging a surgical tool with a tool drive of a surgical robotic system, the method comprising: activating an actuator of a tool drive that is mounted to a surgical robotic arm to rotate a drive disk in the tool drive, wherein the drive disk is to be mechanically engaged with a tool disk in a surgical tool, by activating the actuator to rotate the drive disk at a first velocity until the actuator or the drive disk reaches within a threshold angle away from a reference position at which

point the actuator is activated to rotate the drive disk at a second velocity lower than the first velocity; while the actuator is activated, causing the rotation of the drive disk, monitoring one or more operating parameters of the actuator; and detecting when the drive disk becomes mechanically engaged with the tool disk based on monitoring the one or more operating parameters.

2. The method of claim 1, wherein detecting when the drive disk becomes mechanically engaged with the tool disk comprises: determining that the one or more operating parameters satisfies a condition.

3. The method of claim 2, wherein the operating parameters comprises torque, motor current, motor velocity, or a combination thereof.

4. The method of claim 2, wherein determining that the one or more operating parameters satisfies the condition comprises: determining that measured torque of the actuator exceeds a threshold torque and measured velocity of the actuator is below a threshold velocity, wherein the threshold torque is greater than friction between the drive disk and the tool disk during rotation without mechanical engagement, and wherein the threshold velocity is zero.

5. The method of claim 2, wherein the reference position is a factory determined position stored in an information storage unit of the surgical tool and that corresponds to a hard stop or physical constraint against further rotation of the tool disk and wherein the condition represents the actuator reaching the hard stop or physical constraint.

6. The method of claim 5, wherein the tool disk is winding a cable, and wherein the hard stop or physical constraint comprises a movement constraint caused by a second drive disk in the tool drive that is also winding the cable or holding a position of the cable.

7. The method of claim 1, wherein the one or more operating parameters are collected by one or more sensors.

8. The method of claim 1, wherein the drive disk and the tool disk comprise one or more pairs of coupling features wherein each pair of coupling features becomes interlocked to each other to rotate as one, when mechanical engagement is achieved.

9. The method of claim 1, further comprising: detecting that another drive disk in the tool drive has failed to mechanically engage with another tool disk; and generating a notification for an operator of the surgical robotic system indicating unsuccessful engagement.
